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A gait analysis of simulated knee flexion contracture to elucidate knee-spine syndrome

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Abstract

Knee flexion contracture influences the physiological movements in lower extremities and may cause the kinematic changes of the trunk. Our purpose was to investigate static and dynamic changes in trunk kinematics with simulated knee flexion contracture. Ten healthy females averaged 62 years participated in our study. Unilateral knee flexion contractures of 15° and 30° were simulated with a knee brace. Relaxed standing and level walking were measured at our laboratory using a motion analysis system which consisted of five cameras, a force plate, and thirteen retro-reflective markers. Three-dimensional trunk kinematics and vertical knee forces (% Body Weight) with the contractures were compared with those without the contracture. The 15° contracture did not significantly change trunk kinematics. However, the 30° contracture significantly changed the kinematics in each of the following planes. In the coronal plane, the trunk tilted to the contracture side in standing and walking. In the sagittal plane, posterior inclination of the pelvis in standing significantly increased. In addition, anterior inclination of the trunk and pelvis during walking significantly increased. In the axial plane, trunk rotation to the unaffected side significantly decreased during walking. The vertical knee force in the contracture limb decreased, being accompanied by the increase of the force in the unaffected limb during standing and walking. Results of our study suggest that knee flexion contracture significantly influences three-dimensional trunk kinematics during relaxed standing and level walking, and will lead to spinal imbalance. These facts may explain the onset of the ''Knee-Spine Syndrome''.
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Keywords: Simulated knee flexion contracture; Trunk kinematics; Standing; Gait; Knee-spine syndrome

1. Introduction

Severe knee osteoarthritis often causes a flexion contracture, which affects daily activities [1]. In addition, the contracture influences the physiological movements in other body parts such as the hip and ankle joints [2,3], as well as the spine [4]. Previous studies demonstrated the effect of knee flexion contracture using gait analysis [5,6]. Cerny et al. investigated the effect of unilateral knee flexion contracture on gait using knee contracture simulation and showed that velocity and stride length significantly decreased [2]. Perry et al. examined the knee joint forces required to stabilize the flexed knee during weight bearing using a cadaveric lower extremity. They concluded that a rapid rise in the quadriceps force was required to stabilize the knee between 15° and 30° of flexion [3]. In addition, Potter et al. evaluated standing balance in simulated unilateral and bilateral knee flexion contracture, using a knee brace [7]. They found that the center of pressure moved anteriorly as the bilateral knee flexion contracture increased. It also shifted towards the non-contracture side with unilateral knee flexion contracture. However, to our knowledge, most of these studies concentrated on the kinematics and kinetics of the lower extremities. Only a few papers demonstrated the relationship between knee flexion contracture and trunk kinematics. In these studies, limitation in knee extension was related to a decrease in lumbar lordosis on the sagittal plane radiographs [4,8]. Clinically, back pain is present in 54.6% of patients with knee osteoarthritis, and is more common than in normal population

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[9]. Furthermore, back pain appears to be associated with the decrease of lumbar lordosis, progression of degenerative scoliosis, and progression of rotational deformity [10]. Although it has been demonstrated that back pain and the decrease of lumbar lordosis are related to knee flexion contracture, the relationship between the knee and the spine in the coronal and axial planes has not been investigated [4,8].

We hypothesized that the knee flexion contracture would influence three-dimensional kinematics of the trunk and pelvis. The purpose of the current study was to investigate static and dynamic changes in trunk kinematics of older females with simulated knee flexion contracture during standing and walking.

2. Materials and methods

2.1. Subjects

Ten healthy women, 60–64 years of age (average 62), participated in the current study. Subjects had no history of lower extremity surgery and no pain around the knee. All the subjects provided informed consent and the study was approved by our institution. All methods and procedures were also approved by our institution's ethics committee.

2.2. Gait analysis system and marker locations

The subjects were tested at our gait laboratory using a motion analysis system which consisted of five cameras (120 frames/s; Pro-reflex, Qualisys, Sweden) and a force plate (Frequency 600 Hz, Type 4060-10, Bertec, Columbus, OH, USA). Sample frequency for the force plate was synchronized to the camera sampling rate (120 Hz). Thirteen retro-reflective markers were placed as follows; six markers bilaterally (on the acromion, anterior and posterior superior iliac spines), seven markers on the side to be measured only (lateral epicondyle of the humerus, lateral aspect of the iliac crest, greater trochanter, lateral joint line of the knee, lateral malleolus, lateral aspect of the calcaneus, and head of the fifth metatarsal) (Fig. 1) [11,12].

2.3. Contracture simulation and testing procedures

Unilateral knee flexion contractures of 15° and 30° were simulated for each subject using a hard knee brace (G II Rehabilitation Brace, ALCARE, Tokyo, Japan). In this study, the right limb was selected for the contracture simulation. All the subjects were asked to stand relaxed and to walk on a level floor about 10 m at their preferred speed, with or without the contracture simulation. The walking trials included warm ups, and two trials for each of three conditions were measured until clear contact with the force plate was achieved. The measurements for each leg were undertaken independently as follows; first, the measurement of the contracture limb during relaxed standing (placing one foot on a force plate) and walking trials without the contracture (with brace and no restriction of knee extension) were carried out. The measurements of the contracture limb were then repeated with the simulated flexion contractures. Thereafter, the measurements of the contralateral side were carried out during the same procedure with or without the contracture.

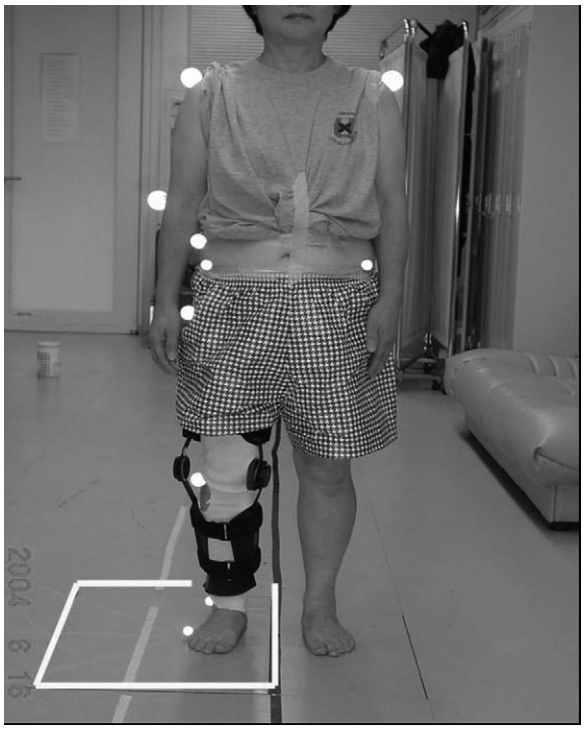


Fig. 1. Relaxed standing and level walking, placing one foot on a force plate.

2.4. Evaluation of the shoulder and trunk motion and knee kinetics

In the coronal plane, the shoulder tilting angle was defined by the height difference in right and left acromions, and the pelvic tilting angle was defined by the height difference in the right and left anterior superior iliac spines (Fig. 2A-a, b). To evaluate trunk tilting in the coronal plane, the shoulder–pelvis bending angle was defined as the angle between the shoulder girdle line (right–left acromion line) and the pelvic line (right–left anterior superior iliac spine line) (Fig. 2A-a, b). To evaluate trunk motion in the sagittal plane, the anterior inclination of the trunk was defined by the slope linked to the contracture side markers on the acromion and iliac crest. Inclination of the pelvis was evaluated by the slope linked to the contracture side markers on the superior anterior iliac spine and superior posterior iliac spine (Fig. 2B-c, d). To evaluate trunk rotation in the axial plane, shoulder–pelvis rotation angle was defined as the angle between the shoulder girdle line and the pelvic line in the axial plane (Fig. 2C-e) [11].

The vertical knee force was calculated with the inverse dynamics approach [13]. The long axis of the tibia was defined as superior–inferior axis, and the external force parallel to this axis was defined as the vertical knee force. The force was normalized to percent body weight of the subjects (%BW). We also evaluated walking velocity (m/s) in each condition.

2.5. Statistical analysis

We evaluated tilting angles of the shoulder and pelvis, anterior inclinations of the trunk, and pelvis, shoulder–pelvis bending angle, shoulder–pelvis rotation angle and bilateral vertical knee force. Maximum of these values during relaxed standing and during

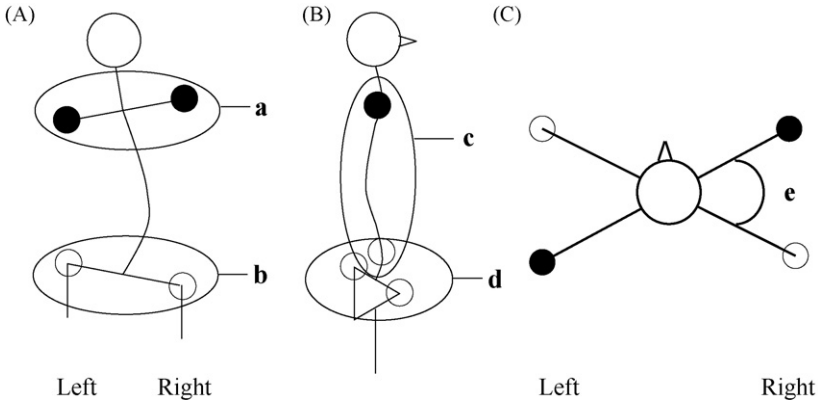


Fig. 2. Measurements of trunk kinematics in each plane: (●) acromion; (○) pelvis. (Right: contracture side, left: non-contracture side.) (A) Coronal plane. a: Shoulder tilting angle; b: pelvic tilting angle. Angles between a and b: shoulder–pelvis bending (P → A view. This illustration shows left bending). (B) Sagittal plane. c: Anterior inclination of trunk; d: anterior inclination of pelvis. (C) Axial plane. e: Shoulder–pelvis rotation (this illustration shows right rotation).

level walking were analyzed as static and dynamic changes due to the contractures, respectively. Shoulder–pelvis bending angle and shoulder–pelvis rotation angle showed a two-peak pattern during each gait cycle, and thus the maximum values on right and left sides were evaluated [11].

An average value of two trials in these parameters during relaxed standing or level walking for each simulation or non-simulation was calculated. Statistical difference between the data with the contractures (15° and 30°) and those without the contracture was evaluated using two-tailed repeated measures of analysis of variance (ANOVA). The SPSS® Version 11 for Microsoft Windows (Chicago, IL) was used for statistical analysis. After a significant P-value (<0.05) was determined, a post hoc Bonferroni correction was performed to compare selected mean values, and P-values of <0.05 were considered as significant.

3. Results

3.1. Static change in relaxed standing (Table 1)

On the whole, the 15° contracture did not significantly change the trunk and pelvic posture, except for pelvic tilt and left bending of the trunk.

In the coronal plane, as the simulated contracture angle increased, the shoulder tilted more to the contralateral side and the pelvis inclined more to the contracture side. In the sagittal plane, anterior inclination of the trunk was not significantly different among different contracture conditions, while anterior inclination of the pelvis significantly decreased with the 30° contracture. In the axial plane, the shoulder–pelvis rotation angle was not significantly changed.

There was remarkable change in the loading condition of both limbs with the 30° contracture. Vertical forces (%BW) of the knee with contracture were as follows; 41.9 ± 2.8 at 0°, 40.7 ± 3.1 at 15°, and 24.0 ± 2.6 at 30° ($P < 0.01$ between 0° and 30°). Those of the contralateral knee were as follows; 41.7 ± 3.0 at 0°, 45.9 ± 1.4 at 15°, and 61.1 ± 5.1 at 30° ($P < 0.01$ between 0° and 30°). These results indicated that the center of gravity shifted towards the unaffected side.

3.2. Dynamic change in level walking (Table 2)

Similarly, the 15° contracture did not significantly change the trunk and pelvic kinematics during walking, except for left shoulder–pelvis rotation.

Kinematic changes in the coronal plane of a representative case with no flexion contracture (0°) and with each simulation (15° and 30°) are shown in Fig. 3A. The shoulder tilted to the contracture side and the pelvis tilted to the contralateral side immediately after the heel-strike of the contracture side foot. On the other hand, the shoulder tilted to the contralateral side and the pelvis to the contracture side immediately after the heel-strike of the unaffected side foot. There was no difference in shoulder and pelvic coronal movement after heel-strike of the unaffected foot. During restriction of knee extension, significant kinematic changes in the coronal plane occurred immediately after the heel-strike of the 30°-flexed knee side. Thus, asymmetrical trunk lateral bending occurred with the 30° contracture, being more pronounced on the side of the knee contracture. In the sagittal plane, the timing of peak values of trunk and pelvic anterior inclination differed for each subject during walking. However, peak angles of the trunk and pelvic anterior inclination during walking significantly increased with the 30° contracture.

Kinematic changes in the axial plane of a representative case with no flexion contracture (0°) and with each simulation (15° and 30°) are shown in Fig. 3B. The shoulder rotated to the contralateral side while the pelvis rotated to the contracture side immediately after the heel-strike of the contracture side foot. On the contrary, the shoulder rotated to the contracture side and the pelvis rotated to the contralateral side immediately after the heel-strike of the unaffected side foot. There was no difference in the rotation after heel-strike of the unaffected foot. During restriction of knee extension, significant kinematic changes occurred immediately after the heel-strike of the 30°-contracture knee side. As a result, contracture side shoulder–pelvis rotation (which meant right rotation in our study;

Table 1
Each data in relaxed standing (mean value ± standard deviation)

	Simulated angles of right knee			
	0°	15°	30°	*P*-value[a]
Shoulder tilt (°) #1	0.10 ± 0.30	−1.43 ± 0.73	−2.03 ± 0.58**	0.04
Pelvic tilt (°) #1	−0.39 ± 0.23	0.58 ± 0.17*	1.95 ± 0.44**	0.00015
Left shoulder–pelvis bending (°)	−0.49 ± 0.40	2.01 ± 0.71*	3.98 ± 0.81**	0.0027
Anterior inclination of pelvis (°)	13.1 ± 1.63	12.5 ± 1.63	11.4 ± 1.59**	0.0062
Anterior inclination of trunk (°)	0.26 ± 0.55	0.61 ± 0.66	0.59 ± 0.75	0.8
Shoulder–pelvis rotation (°) #2	−1.40 ± 0.58	−1.74 ± 1.07	0.37 ± 0.69	0.11
Right knee vertical force (%BW)	41.9 ± 2.8	40.7 ± 3.1	24.0 ± 2.6**	0.000046
Left knee vertical force (%BW)	41.7 ± 3.0	45.9 ± 1.4	61.1 ± 5.1**	0.0000014

#1 Positive values were obtained when right side was lower than left side.
#2 Positive values were obtained when right acromion was more anterior than right superior anterior iliac spine.
[a] Values obtained using repeated measures of analysis of variance.
* $P < 0.05$ between 0° and 15°.
** $P < 0.05$ between 0° and 30°.

Fig. 2C-e) was not significantly different, while the unaffected side rotation (which meant left rotation in our study) significantly decreased with the 30° contracture. Asymmetrical trunk motion was also seen in the axial plane.

Similarly to relaxed standing, the peak vertical knee force on the contracture limb decreased, and those on the contralateral limb significantly increased with the 30° contracture. Walking velocities were as follows; 1.17 ± 0.19 m/s at 0°, 1.12 ± 0.26 m/s at 15°, and 0.98 ± 0.27 m/s at 30° contracture. Walking velocity significantly decreased with the 30° contracture ($P < 0.05$ between 0° and 30°).

4. Discussion

The results of this study support our hypothesis that simulated knee flexion contracture leads to three-dimensional changes in trunk kinematics. To our knowledge, the current study was the first to evaluate both static and dynamic influences of simulated knee flexion contracture on trunk kinematics using motion analysis. As the knee contracture increased, the loading on the flexed knee side reduced, while loading on the unaffected knee increased during both standing and walking.

Although we did not directly measure the changes in the spine, the postural changes in the trunk during relaxed standing suggest that the trunk bends away from the contracture side in the coronal plane when a severe knee flexion contracture is simulated. In addition, posterior inclination of the pelvis significantly increased with the 30° contracture. We thus confirmed the findings of the previous studies that indicated the relation between knee flexion contracture and decrease in lumbar lordosis and pelvic posterior inclination by radiographs during standing [4,8].

In the coronal plane, the trunk bent away from the knee contracture side during walking. The increase in the anterior inclination of the pelvis with the 30° contracture would suggest that the center of mass moved more anteriorly due to the contracture. Trunk movement in the axial plane was also affected. The flexion contracture led to asymmetrical rotational movements of the trunk. In particular, trunk rotation towards the contracture side (which meant right shoulder–pelvis rotation in our study; Fig. 2C-e) showed a trend towards an increase, although this was not significant. The rotation towards the unaffected side (which meant left

Table 2
Each data in walking trials (mean value ± standard deviation)

	Simulated angles of right knee			
	0°	15°	30°	*P*-value[a]
Walking velocity (m/s)	1.17 ± 0.19	1.12 ± 0.26	0.98 ± 0.27**	0.00068
Right shoulder–pelvis bending (°)	4.53 ± 1.59	3.81 ± 1.91	1.83 ± 2.30**	0.01
Left shoulder–pelvis bending (°)	4.49 ± 1.23	4.94 ± 1.36	4.51 ± 1.57	0.79
Anterior inclination of trunk (°)	0.20 ± 1.76	0.66 ± 1.73	3.07 ± 1.58**	0.0011
Anterior inclination of pelvis (°)	16.7 ± 1.19	17.6 ± 1.17	19.2 ± 1.21**	0.00036
Right shoulder–pelvis rotation (°)	5.47 ± 1.37	5.50 ± 1.20	6.66 ± 1.52	0.31
Left shoulder–pelvis rotation (°)	7.04 ± 1.24	6.16 ± 1.35*	4.33 ± 2.07**	0.039
Right knee vertical force (%BW)	107.5 ± 3.7	109.3 ± 2.3	99.1 ± 2.9**	0.015
Left knee vertical force (%BW)	105.9 ± 2.9	112.1 ± 2.6	111.2 ± 3.2**	0.007

[a] Values obtained using repeated measures of analysis of variance.
* $P < 0.05$ between 0° and 15°.
** $P < 0.05$ between 0° and 30°.

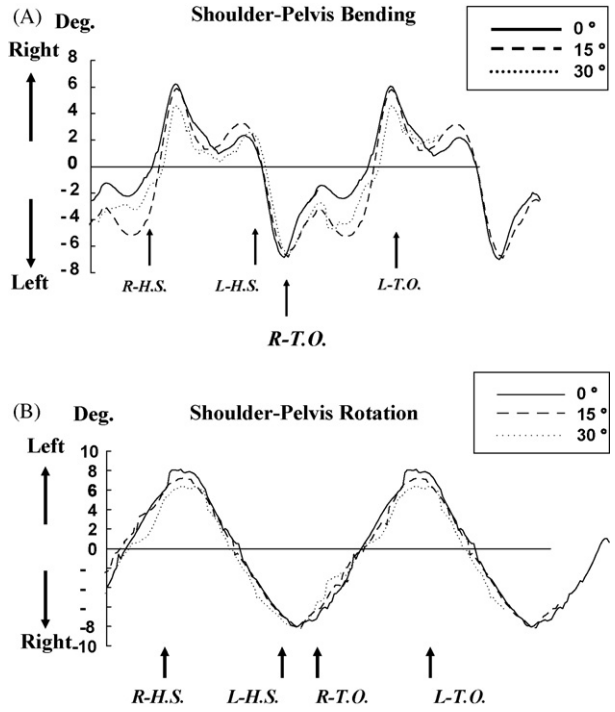


Fig. 3. (A) A representative kinematic changes in the coronal plane during walking with or without simulation. Immediately after the right heel-strike, trunk bent convexly to the left (right bending). R: right, L: left, H.S.: heel-strike, T.O.: toe off. (B) A representative kinematic changes in the axial plane during walking with or without simulation. Immediately after the right heel-strike, trunk rotated towards the left (left shoulder–pelvis rotation). R: right, L: left, H.S.: heel-strike, T.O.: toe off.

shoulder–pelvis rotation in our study) significantly decreased (Table 2). These dynamic changes in the trunk motion during walking could lead to changes of the spinal alignment in the coronal, sagittal and axial plane, and possibly contribute to the development of spinal deformity. From our results, we speculate that unilateral knee flexion contracture can contribute to the onset of spinal disorders, and these facts may explain the onset of the ‘‘Knee-Spine Syndrome’’.

Hip-spine syndrome was first described by Offierski and MacNab [14]. They revealed that lumbar lordosis and sacral inclination were related to the degree of hip extension. Thereafter, hip-spine syndrome was examined by several researchers [15–17]. In contrast to the hip-spine syndrome, few studies have explored the knee-spine syndrome and examined the correlation between knee joint pain and low back pain [4,8,9,18]. These studies limited their analysis to the sagittal plane and did not assess the three-dimensional changes of the trunk kinematics. The present study successfully demonstrated that knee flexion contracture dynamically changes the trunk kinematics. A previous study on simulated leg length discrepancy of three centimeters demonstrated asymmetrical gait [11]. Similarly, we have shown that the influence on the trunk kinematics depends on the severity of the knee flexion contracture. Our results suggest that a flexion contracture of less than 15° would not cause any adverse effect to the trunk.

Mündermann et al. suggested that patients with knee osteoarthritis shifted their body weight rapidly to the osteoarthritic knee side following heel-strike in the coronal plane to reduce the knee adduction moment [19]. However, patients with severe knee flexion contracture were excluded from their study. On the contrary, the present study showed that subjects with severe knee flexion contracture did not shift their body weight to the knee contracture side. These facts suggest that trunk movement in patients with knee osteoarthritis becomes more complicated depending on the degree of knee flexion contracture or varus deformity.

Several limitations should be noted in this study. First, skin markers were not placed on the spinous processes to measure the change in the spinal alignment. This is because of difficulties in both identifying and attaching the makers on the spinous process in the lumbar spine. Furthermore, it could be argued that our findings are based on the influence of leg length discrepancy during gait, caused by the unilateral knee flexion contracture [11]. In fact, unilateral knee flexion contracture is usually seen in cases of osteonecrosis or post-traumatic arthritis, while the majority of patients are affected bilaterally. Unilateral knee flexion contracture is a rarer occurrence. Nevertheless, remarkable changes were observed in our study both in the limb loading and the trunk kinematics. Therefore, our results offer useful information when considering the treatment of such patients.

5. Conclusion

Unilateral knee flexion contracture dramatically increased weight bearing in the contralateral knee and decreased in the ipsilateral knee. The current study confirmed our hypothesis that a knee flexion contracture would lead to three-dimensional changes in trunk kinematics.

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Conflict of interest

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